Stem-and-leaf diagrams

1 This stem-and-leaf diagram records the ages of customers at a beachside cafe last Sunday. Use the stem-and-leaf diagram to complete the following frequency table:

	Leaf
1	1 5 9
2	2 3 5 6 7 9
3	0 1 2 4 6 7
4	1 3 4 6 8 9
5	1 4 7
6	2 4 5 6 7

Key: $5|2 = 52$

Age	Frequency
$10 - 19$	
$20 - 29$	
$30 - 39$	
$40 - 49$	
$50 - 59$	
$60 - 69$	

2 An online booking service allows customers to book tickets to see performers in concert. Over the last month the service has been tracking how many tickets are sold in the first five seconds after they become available, and their results are displayed in the given stem-and-leaf diagram:

a How many concert events does this data set represent?

b Find the highest number of tickets sold in the first five seconds.

c Find the difference between the highest and lowest number of tickets sold in the first five seconds.

	Leaf
1	0 0 1 2 2 3 4 8
2	0 1 5 7 8 9
3	0 2 6 8
4	
5	6

Key: $1|2 = 12$ tickets sold

3 The following stem and leaf diagram shows the age of people who enter through the gates of a theme park in the first ten minutes:

a How many people passed through the gates in the first ten minutes?

b State the age of the youngest person.

c State the age of the oldest person.

d Find the age difference between the youngest and oldest person.

	Leaf
1	1 2 3 3 4 4 6 6 7
2	0 2 4 9 9
3	2 5 5 5
4	
5	3

Key: $1|2 = 12$ years old

4 A city council selected a number of house[illegible]andom. They determined the fastest travel time (in minutes) from each house to the nearest hospital, and recorded the following results:

$$25, 37, 16, 27, 27, 35, 21, 18, 19, 49, 14, 19, 31, 42, 18$$

Represent this data as an ordered stem-and-leaf diagram.

5 Consider the given stem-and-leaf diagram:

Find the mean of this data set.

	Leaf
1	2 6 8 9
2	0 2 9
3	0 2

Key: $1|2 = 12$

6 Consider the given stem-and-leaf diagram:

Find the median of this data set.

	Leaf
1	2 9
2	4 6 7
3	0 4 5 6
4	3 5 7 9
5	0 2
6	5
7	3 9
8	4 5

Key: $5|2 = 52$

7 The level of mercury in 40 different water samples was tested and recorded in the following stem-and-leaf diagram:

a State the highest reading.

b Find the 8th lowest reading.

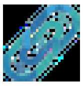
c How many readings of 100 or higher were recorded?

d The level of mercury is deemed to be safe if the reading is 103 or lower. What fraction of the readings are safe?

	Leaf
9	3 3 3 3 6 7 8 8 9 9
10	0 0 3 3 4 4 5 5 6 7 9
11	2 3 4 4 5 6 8 9 9
12	0 1 1 1 2 2 3 5 6 7

Key: $1|2 = 12$ units

8 Roxanne is using a precision telescope to measure the size of craters (in km) closest to Olympus Mons on the surface of Mars. She records her findings in the given stem-and-leaf diagram:



	Leaf
1	1 2 3 6 7 8 8
2	1 3 4 5 5 5
3	0 1 1 2 3 4 4 5 7 8 8
4	0

Key: $1|2 = 12\text{ km}$

a Find the median crater size.

b Find the range of crater sizes.

c Roxanne finds that the sum of all the crater sizes is 661. Find the mean crater size, correct to two decimal places.

9 The given stem-and-leaf diagram shows the ages of 20 employees in a company:

	Leaf
2	0 1 1 2 8 8 9
3	0 2 4 8 8
4	1 1 1 2 5
5	3 4 8

Key: $1|2 = 12$ years old

a Find the number of employees in their 30s.

b State the age of the oldest employee.

c State the age of the youngest employee.

d Find the median age of the employees.

e Find the modal age.

10 A cyclist measured his heart rate immediately after finishing each event in which he competed. The results are recorded in the following stem and leaf diagram:

	Leaf
16	1 4 4 4 6 8 8
17	0 0 4 6 8
18	0 2 3 3
19	1 2 3 4 7 7 8

Key: $1|2 = 12$ beats per minute

a Calculate the difference between his slowest and fastest heart rate.

b Calculate the mean heat rate of the cyclist, to the nearest whole number.

c Find the median heart rate.

d Find the modal heart rate.

e For each extra heartbeat per minute, it takes 7 seconds longer for him to recover. How much longer would it take to recover when he finishes with the fastest heart rate, compared to when he finishes with the slowest heart rate?